HDI Predictor: A Comprehensive Measure of Well-Being

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
GitHub	https://github.com/karnayanaBala/HDI-Predictor-ML-Project

Project Description:

The HDI Predictor is a Machine Learning web application that predicts the Human Development Index (HDI) score and development category for any country based on 4 key indicators: Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) Per Capita.

Built using Python, Flask, Scikit-learn, Pandas, NumPy, Matplotlib, and Seaborn. The Linear Regression model is trained on Kaggle HDI dataset (195 countries) and achieves R2 accuracy of 97.73%. Deployed locally using Flask at http://127.0.0.1:5000

Scenarios

Scenario 1: Predicting Very High Human Development

User enters developed country values (Life Expectancy=80, Expected Schooling=16, Mean Schooling=13, GNI=60000). Model predicts Very High HDI score of 0.92.

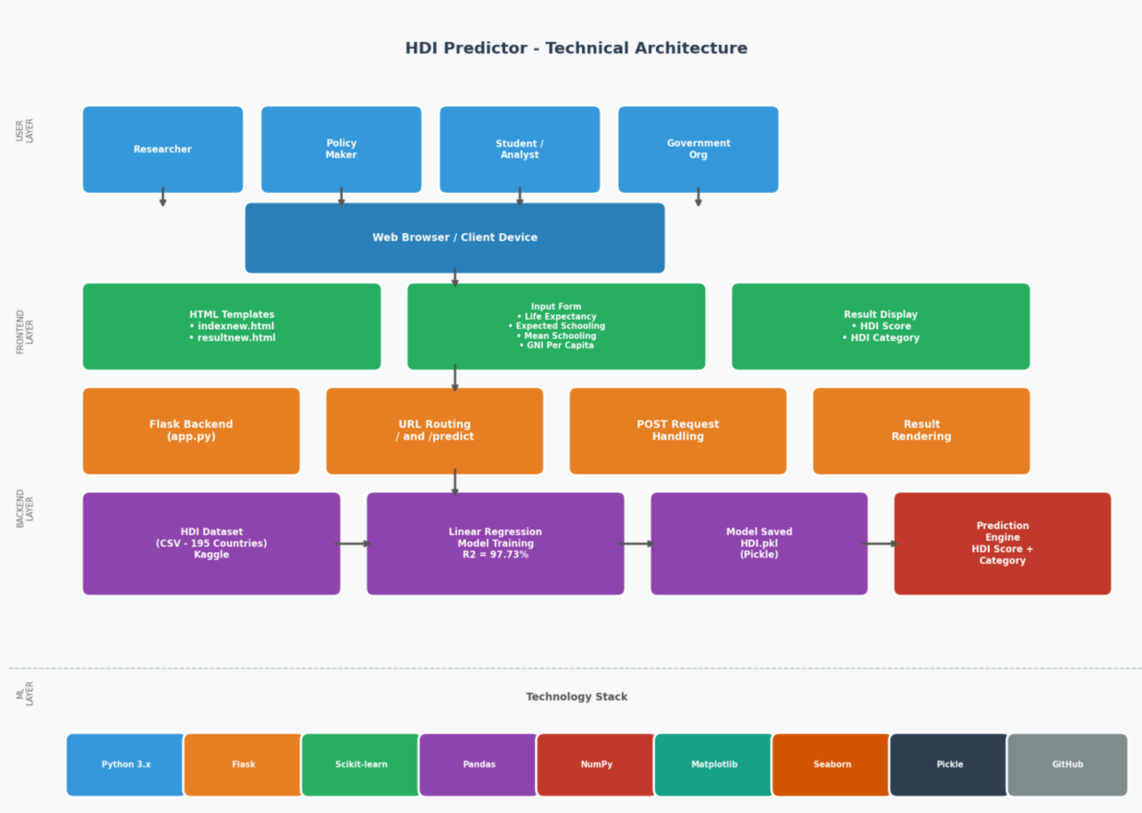
Scenario 2: Identifying Development Gaps

Policymaker inputs mid-range values (Life Expectancy=70, Expected Schooling=12, Mean Schooling=7, GNI=8000). Model returns Medium HDI score of 0.65.

Scenario 3: Assessing Low Development Countries

Researcher evaluates low indicators (Life Expectancy=55, Expected Schooling=8, Mean Schooling=3, GNI=1500). Model predicts Low HDI score of 0.48.

Technical Architecture:



Description: The HDI Predictor uses a three-tier architecture:

- User Layer: Researchers, Policy Makers, Students, Government Organizations via web browser
- Frontend Layer: HTML/CSS templates (indexnew.html, resultnew.html) with Jinja2
- Backend Layer: Flask app.py - URL routing (/ and /predict), POST handling, result rendering
- ML Layer: CSV dataset → Linear Regression → HDI.pkl → Prediction Engine (R2=97.73%)

Pre-requisites:

- Python 3.x - https://www.python.org/downloads/
- Flask - pip install flask
- Scikit-learn - pip install scikit-learn
- Pandas, NumPy - pip install pandas numpy
- Matplotlib, Seaborn - pip install matplotlib seaborn
- Kaggle HDI Dataset - https://www.kaggle.com/datasets/iamsouravbanerjee/human-development-index-dataset
- GitHub - https://github.com/karnayanaBala/HDI-Predictor-ML-Project

Project Workflow:

Milestone 1: Data Collection and Environment Setup

- Install numpy, pandas, matplotlib, seaborn, scikit-learn, flask
- Download HDI dataset from Kaggle (195 countries, 880 columns)
- Load with Pandas and select 7 key columns

Milestone 2: Data Visualization and EDA

- Bar plot - HDI Category distribution
- Histogram with KDE - HDI Score distribution
- Correlation Heatmap - relationships between features

Milestone 3: Data Preprocessing

- Drop null values (195 → 191 clean rows)
- Label Encoding - HDI Category to numbers
- Feature Selection - X (4 features), y (HDI Score)

Milestone 4: Model Building and Saving

- Train-Test Split 75/25 (143 train, 48 test)
- Train Linear Regression model
- R2 Score = 97.73% accuracy
- Save model as HDI.pkl using Pickle

Milestone 5: Flask Web Application

- Create app.py with / and /predict routes
- Create indexnew.html with 4 input fields
- Create resultnew.html for prediction results
- Test at http://127.0.0.1:5000 - working correctly

Conclusion:

The HDI Predictor successfully demonstrates a complete ML pipeline from data collection to web deployment. R2 accuracy = 97.73%. Developed by Bala Karnayana, Mohan Babu University, Tirupati. Internship: 25 May 2026 - 8 July 2026. GitHub: https://github.com/karnayanaBala/HDI-Predictor-ML-Project